

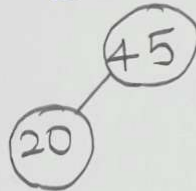
CO3 - AT2

1) Construct a Binary search tree (BST) using dataset 45, 20, 60, 10, 30, 50, 70, 25, 35

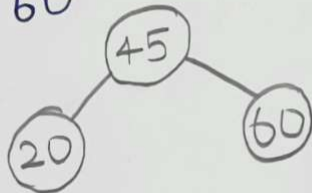
• Insert 45



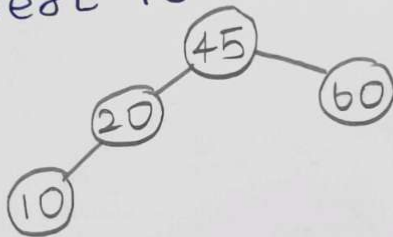
• Insert 20



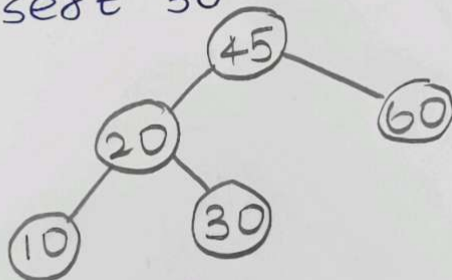
• Insert 60



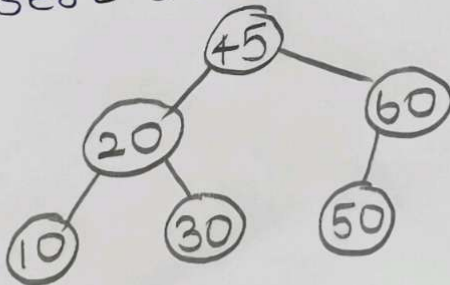
• Insert 10



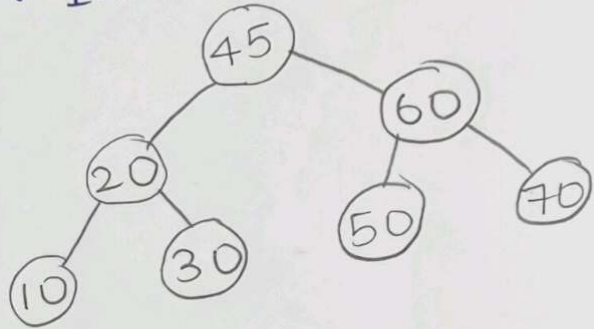
• Insert 30



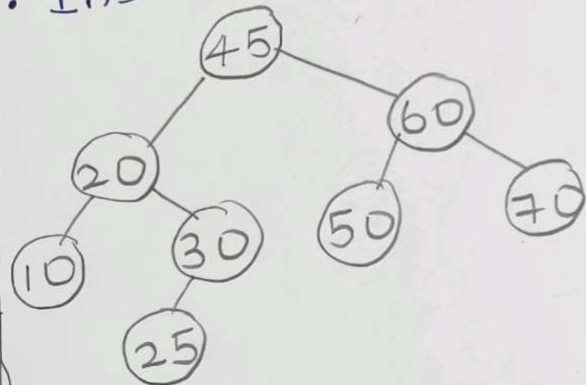
• Insert 50



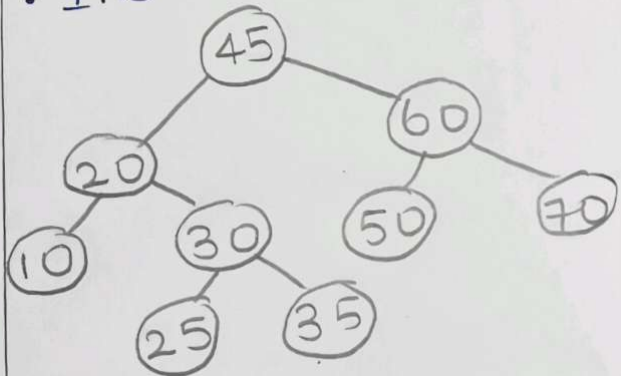
• Insert 70



• Insert 25



• Insert 35

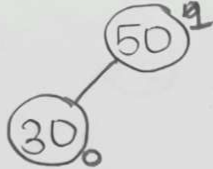


2) Construct an AVL tree using elements 50, 30, 70, 20, 40, 60, 80, perform the required operation, delete node 30.

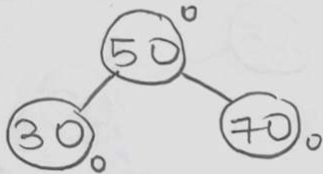
• Insert 50



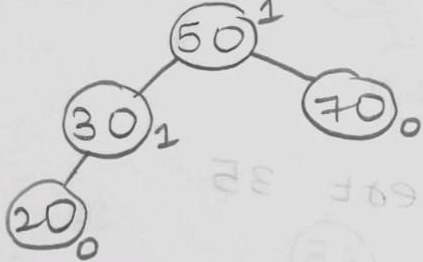
• Insert 30



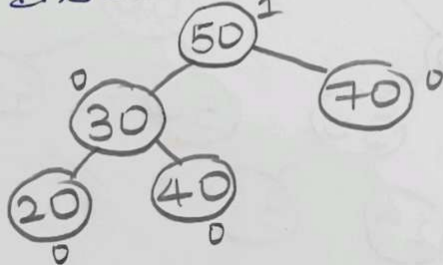
• Insert 70



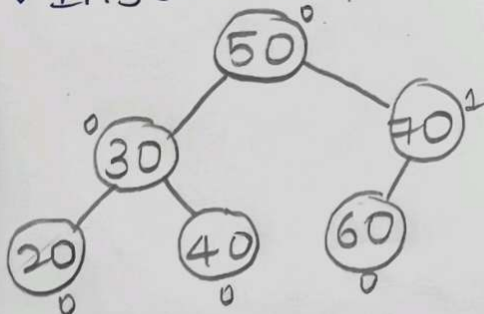
• Insert 20



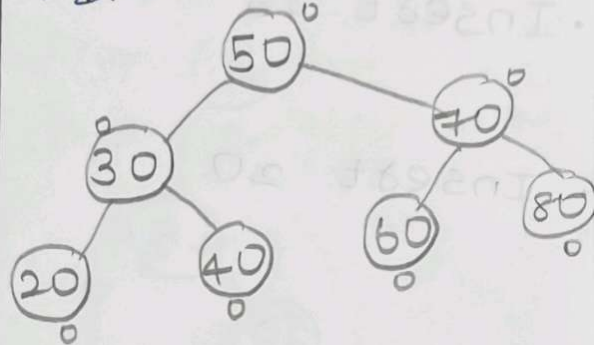
• Insert 40



• Insert 60

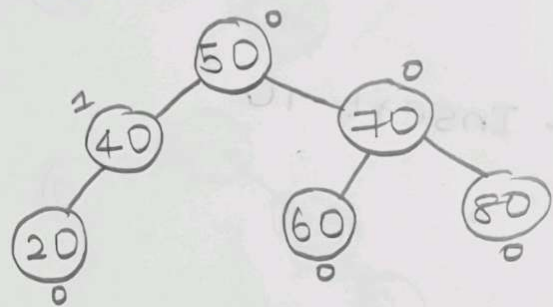


• Insert 80



Balanced

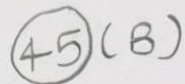
• Delete 30



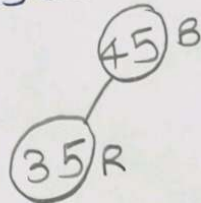
Balanced

3) Construct Red black tree using {45, 35, 55, 25, 40, 50, 60} delete node 35.

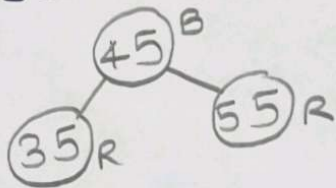
• Insert 45



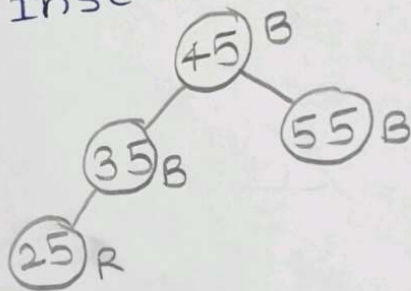
• Insert 35



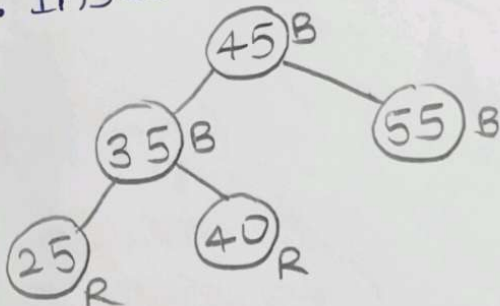
• Insert 55



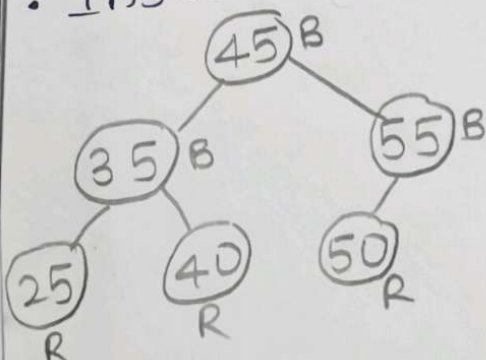
• Insert 25



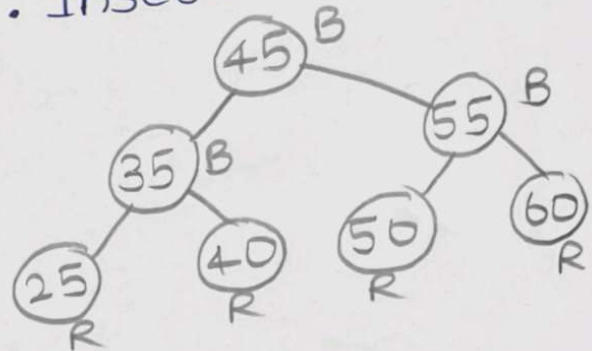
• Insert 40



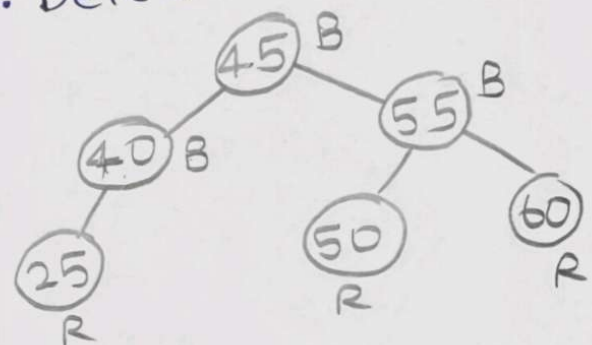
• Insert 50



• Insert 60



• delete 35

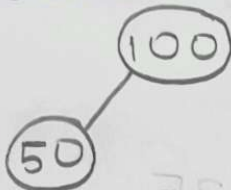


4) Develop a splay tree with elements (100, 50, 200, 40, 30, 20), perform search operation for 20, and demonstrate the zig-zig rotations that bring to root.

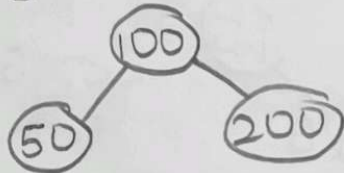
• Insert 100



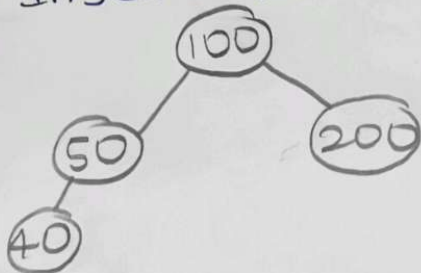
• Insert 50



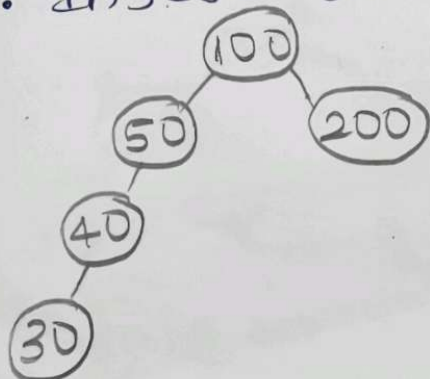
• Insert 200



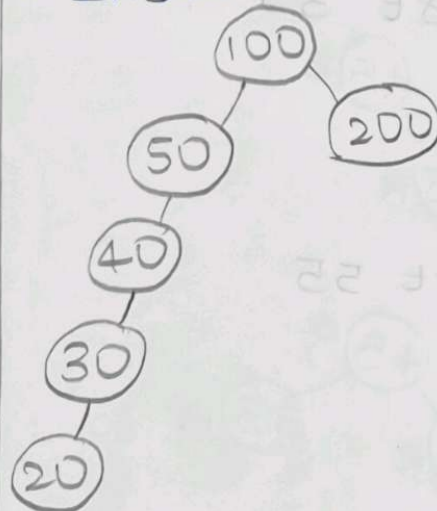
• Insert 40



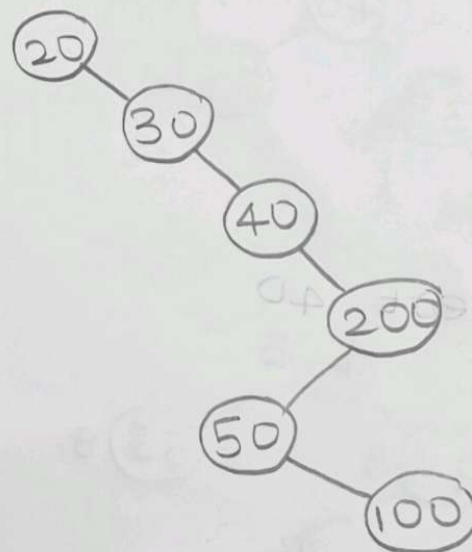
• Insert 30



• Insert 20



→ splay tree:



5) create 2-3-4 tree for dataset [50, 40, 60, 30, 70, 20, 80] perform insertions step by step and show how 4-nodes are split during the process.

- Insert 50
[50]

- Insert 40
[40 | 50]

- Insert 60
[40 | 50 | 60]

- Insert 30
[50]
/ \
[30 | 40] [60]

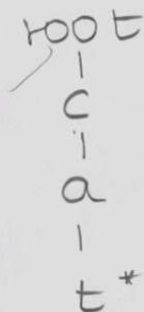
- Insert 70
[50]
/ \
[30 | 40] [60 | 70]

- Insert 20
[50]
/ \
[20 | 30 | 40] [60 | 70]

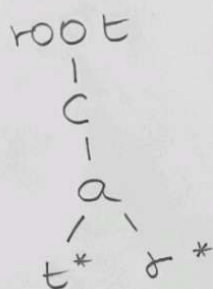
- Insert 80
[50]
/ \
[20 | 30 | 40] [60 | 70 | 80]

b) construct a trie using words {cat, car, care, dog, dot} perform insertion for all words, search for prefix "ca", delete "car" while maintaining trie.

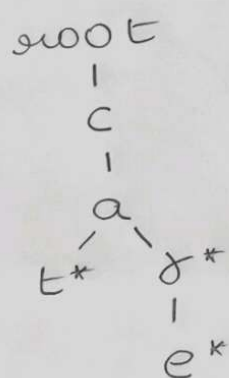
• Insert cat:-



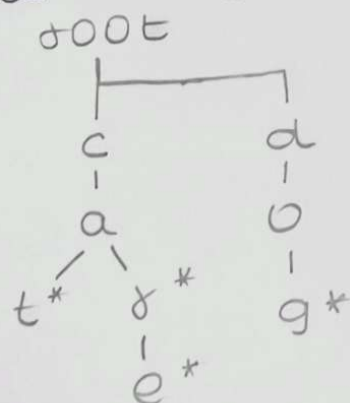
• Insert car:-



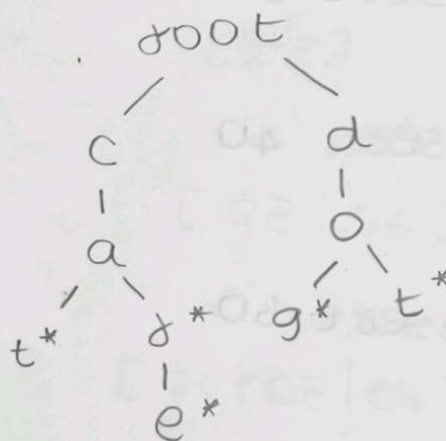
• Insert care:-



• Insert dog:-



• Insert dot:-



• search ca

ca → {cat, car, care}

• Delete car

